

COMPUTER PROGRAMMING

LAB NO: 04

LAB TITLE

Switch Statement, Break and Continue in C++

1.OBJECTIVE

The objective of this lab is to understand the use of the switch statement in C++. Students also learn how the break statement controls program flow and how the continue statement is used inside loops. This lab also introduces simple menu-driven programs using switch statements.

2. THEORY

The switch statement is used when a program needs to select one option from multiple choices based on the value of a variable. It is often used instead of multiple if-else statements when comparing a single variable with several constant values. Each option inside the switch statement is written as a case. The program checks the value of the variable and executes the matching case block. The break statement is used to stop execution inside a case. Without break, the program would continue executing the next cases. The default statement is optional and runs when none of the case values match the given input. The continue statement is used inside loops to skip the current iteration and move to the next iteration.

3. SOFTWARE

- 1.DEV-C++ IDE
- 2.C++ Compiler

4. PROCEDURE / PROCESS

Day Name Using Switch

- 1.Start the program
- 2.Declare an integer variable for day number
- 3.Ask the user to enter a number from 1 to 7
- 4.Use a switch statement to check the number
- 5.Display the corresponding day name
- 6.End the program

```
#include <iostream> // input aur output ke liye library using

namespace std; // standard namespace use karne ke liye int

main() {

    int day; // integer variable jisme day number store hoga cout << "Enter day
    number (1-7): "; // user se day number input lene ka message cin >> day; // user
    se value lena switch(day) // day ke mutabiq case select hoga{ case 1: cout <<
    "Monday"; break; // agar 1 ho to Monday case 2: cout << "Tuesday"; break;
    // agar 2 ho to Tuesday case 3: cout << "Wednesday"; break; // agar 3 ho to
    Wednesday case 4: cout << "Thursday"; break; // agar 4 ho to Thursday case 5:
    cout << "Friday"; break; // agar 5 ho to Friday case 6: cout << "Saturday";
    break; // agar 6 ho to Saturday case 7: cout << "Sunday"; break; // agar 7 ho
    to Sunday default: cout << "Invalid day number";
        // agar 1-7 ke ilawa koi number ho to error message
    }

    return 0; // program end
}
```

Grade Evaluation Using Switch

- 1.Start the program
- 2.Declare a character variable for grade
- 3.Ask the user to enter a grade
- 4.Use a switch statement to evaluate the grade
- 5.Display the corresponding message

6.End the program

```
#include <iostream> // input aur output ke liye library
#include <cctype> // toupper function ke liye library using
namespace std; // standard namespace
```

```
int main() { char grade; // character variable jisme grade store hoga
cout << "Enter your grade (A, B, C, D, F): "; // user se grade lena
cin >> grade; // input lena grade = toupper(grade); // lowercase ko
uppercase me convert karna switch(grade) { // grade ke mutabiq
decision lena case 'A': cout << "Excellent"; // A grade ka result
break; case 'B':
cout << "Very Good"; // B grade ka result
break; case 'C':
cout << "Good"; // C grade ka result
break; case 'D': cout << "Pass"; //
D grade ka result break; case 'F':
cout << "Fail"; // F grade ka result
break; default:
cout << "Invalid Grade"; // agar koi aur input ho
} return 0; // program
end}
```

Unit Conversion Menu Using Switch

- 1.Start the program
- 2.Display a menu of conversion options
- 3.Take user choice
- 4.Ask for input value
- 5.Perform the selected conversion
- 6.Display the result
- 7.End the program.

```
#include <iostream> // input aur output ke liye library using
namespace std; // standard namespace int main() { int choice; //
user ki choice store karne ke liye double value; // value store karne
ke liye (decimal bhi ho sakti hai)
// menu display karna cout << "1 → Kilometers to Meters\n"; cout << "2 → Meters
to Centimeters\n"; cout << "3 → Kilograms to Grams\n"; cout << "4 → Celsius to
Fahrenheit\n"; cout << "Enter your choice: "; cin >> choice; // choice input lena
```

```

cout << "Enter value: "; cin >> value; // value input lena switch(choice) // choice
ke mutabiq conversion hoga { case 1: cout << "Meters = " << value * 1000; //
kilometers ko meters me convert karna break; case 2: cout << "Centimeters = " <<
value * 100; // meters ko centimeters me convert karna break; case 3:
cout << "Grams = " << value * 1000; // kilograms ko grams me convert karna break;
case 4: cout << "Fahrenheit = " << (value * 9/5) + 32; // celsius ko fahrenheit me
convert karna break; default:
cout << "Invalid choice"; // agar galat option h}
return 0; // program end}

```

7. CONCLUSION

In this lab we learned how to use the switch statement for multiple decision making in C++. We also understood the importance of the break statement in controlling program flow. This lab helped us develop simple menu-based programs and improved our understanding of conditional structures.